Mobile and Cellular Communication Laboratory

Theory, MATLAB Simulation Methods, and Worked Results

A laboratory explainer for nine cellular-communications experiments

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Prepared from the supplied MATLAB scripts

Illustrative figures use the scripts' stated default parameters where practical

13 October 2026

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1 Abstract

This report explains nine MATLAB experiments spanning radio propagation, cellular frequency planning, GSM mobility management, and teletraffic engineering. Each chapter connects the implementation to its governing model, defines symbols and units, derives the central equations, describes the figures, and provides representative results at the code's default or clearly stated illustrative settings. The plots are explanatory simulations, not field measurements. Where a plot is an independently generated Python rendering of a MATLAB model, a different pseudo-random number generator may produce a different stochastic trace even with the same numeric seed.

2 How to use this report

The code archive contains nine interactive MATLAB scripts. Unless values are entered at the prompts, each script uses its displayed default. The report refers to those source files by experiment number. Before running a script, verify units, choose physically meaningful inputs, and save the console output and figures with the submitted report. MATLAB functions such as `input`, `sind`, `cosd`, `accumarray`, and local functions are used in the supplied code; they are MATLAB syntax, not separate measured data.

2.1 Unit and notation conventions

- f_c: carrier frequency in hertz (Hz) unless explicitly labelled MHz.
- $c \approx 3 \times 10^8$ m/s: speed of light; $\lambda = \frac{c}{f_c}$: wavelength in metres.
- v: mobile speed in m/s; R: hexagonal cell radius; D: nearest co-channel center spacing.
- dBm is an absolute power referenced to 1 mW; dBi is antenna gain relative to isotropic; dB is a power ratio.
- Traffic in Erlangs is the mean number of simultaneously occupied circuits under the offered load.
- For log-distance propagation, distances must be ratioed in the same unit; do not mix metre and kilometre values without conversion.

2.2 Lab-report procedure and reproducibility

Record the exact prompt values, software version, and random seed; run each MATLAB script; compare printed results with equations below; inspect every generated plot; then discuss deviations from ideal assumptions. This report includes one composite explanatory figure per experiment. These figures illustrate the model, and do not replace MATLAB output when an instructor requires original program output. Experiment 4 uses synthetic subscriber identifiers; they are not live network records. Experiment 5 and the subscriber shadowing in Experiment 6 are random, so traces are seed- and implementation-dependent.

1 Calculation of Doppler Shift and Maximum Mobile Velocity

1.1 Aim

Compute carrier wavelength, maximum and angle-dependent Doppler shift, the maximum velocity allowed by a Doppler limit, and an approximate channel coherence time.

1.2 Theory and derivation

Relative motion changes the apparent carrier frequency. For a plane wave arriving at angle θ measured between the mobile's velocity vector and the incoming propagation direction, the supplied convention is

$$f_{D(\theta)} = \left(\frac{v}{\lambda}\right)\cos(\theta) = \left(v\frac{f_c}{c}\right)\cos(\theta).$$

The maximum magnitude occurs in the radial direction ($|\cos(\theta)| = 1$):

$$f_{\max} = \frac{v}{\lambda} = v\frac{f_c}{c}.$$

At $\theta = 90°$, the first-order radial Doppler is zero; at 180° the sign reverses. Convert speed from km/h to m/s by dividing the numeric value by 3.6. A receiver that tolerates a maximum Doppler magnitude f_{tol} imposes

$$v_{\max} = f_{\text{tol}}\lambda = f_{\text{tol}}\frac{c}{f_c}, \quad v_{\max} = 3.6 f_{\text{tol}}\frac{c}{f_c}.$$

A common rough coherence-time estimate is $T_c \approx \frac{0.423}{f_{\max}}$. It summarizes fading variation timescale; coherence-time definitions depend on the chosen correlation threshold and spectrum model.

1.3 Supplied MATLAB method

`exp1_doppler_shift.m` prompts for f_c, mobile speed, arrival angle, and allowable Doppler. It uses the equations above and displays (i) shift versus angle, (ii) maximum shift versus speed at several carriers, (iii) allowed speed versus carrier for the selected Doppler limit, and (iv) coherence time versus speed on a logarithmic vertical axis.

1.4 Worked default result and interpretation

For $f_c = 900$ MHz, $v = 60$ km/h $= 16.667$ m/s, $\theta = 0°$, and $f_{\text{tol}} = 100$ Hz, $\lambda = 0.3333$ m, $f_{\max} = 50.0$ Hz, and $f_{D(0)} = 50.0$ Hz. The threshold permits $v_{\max} = 33.333$ m/s $= 120$ km/h. $T_c \approx \frac{0.423}{50} = 0.00846$ s $= 8.46$ ms. Higher carrier frequency or speed increases Doppler; the allowed speed for a fixed Doppler ceiling decreases inversely with carrier frequency.

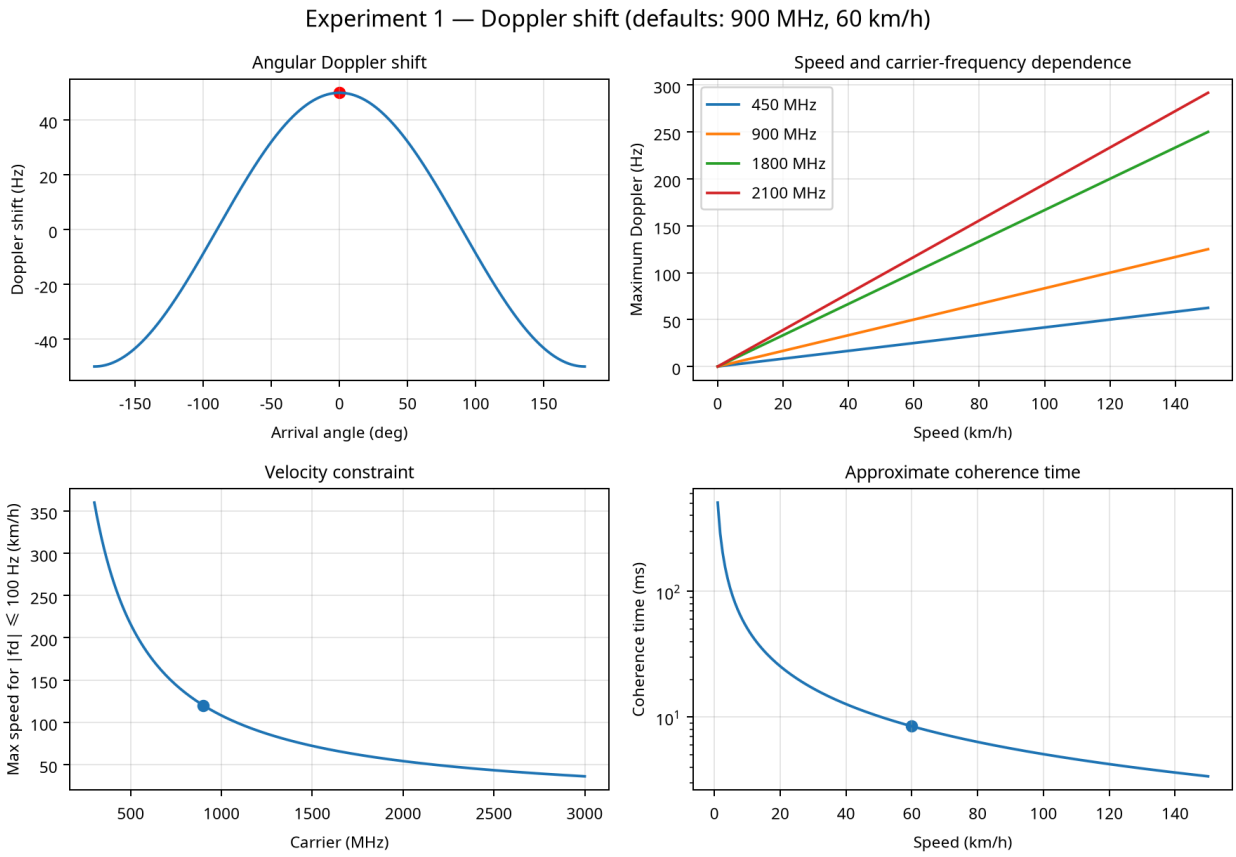


图 1 Doppler shift trends. The marked operating point is 900 MHz and 60 km/h.

1.5 Checks and limitations

The frequency in the code is first converted from MHz to Hz; the angle is treated in degrees. The coherence-time expression uses maximum Doppler, not the selected angle-specific Doppler. It is an approximation for a classical isotropic-scattering setting, not a full channel estimator. Negative Doppler is a signed frequency shift; use magnitude when applying a tolerance.

2 Calculation of Co-channel Reuse Distance and Reuse Ratio

2.1 Aim

Relate a hexagonal reuse cluster to co-channel separation, reuse ratio, an idealized first-tier signal-to-interference ratio, and the fraction of system channels assigned to each cell.

2.2 Theory

A regular hexagonal layout permits cluster sizes

$$N = i^2 + ij + j^2,$$

where i, j are nonnegative integer lattice shifts, not both zero. The standard center-to-center reuse distance is

$$D = R\sqrt{3N}, \quad Q = \frac{D}{R} = \sqrt{3N}.$$

Thus an N-cell reuse cluster assigns approximately one of every N system channels to an individual cell; increasing N separates repeated frequencies further but reduces the per-cell share in a fixed spectrum pool. Under the simplified equal-distance, equal-power, six-first-tier co-channel-interferer model, path-loss exponent n gives

$$\gamma \approx \frac{\left(\frac{D}{R}\right)^n}{6} = \frac{Q^n}{6},$$

$$\gamma_{\mathrm{dB}} = 10\log_{10}\left(\frac{Q^n}{6}\right).$$

This is a planning approximation, not a location-specific SINR prediction: actual interferer distances, shadowing, antenna patterns, sectorization, and load are ignored.

2.3 Supplied MATLAB method and figure

`exp2_cochannel_reuse.m` validates i, j, computes N, D, Q, and SIR, then constructs a coloured hexagonal lattice, highlights six nearest co-channel cells, and plots Q and approximate SIR versus valid cluster sizes. The dashed segments in the lattice mark reuse spacing. The colour classes are a visual reuse-group representation.

2.4 Worked default result

At $i = 2$, $j = 1$, and $R = 1$ km, $N = 4 + 2 + 1 = 7$, $Q = \sqrt{21} = 4.583$, and $D = 4.583$ km. For $n = 4$, $\gamma \approx \frac{4.583^4}{6} = 73.5$, or about 18.66 dB. In a spectrum with a total of C channels, the idealized per-cell allocation is approximately $\frac{C}{7}$.

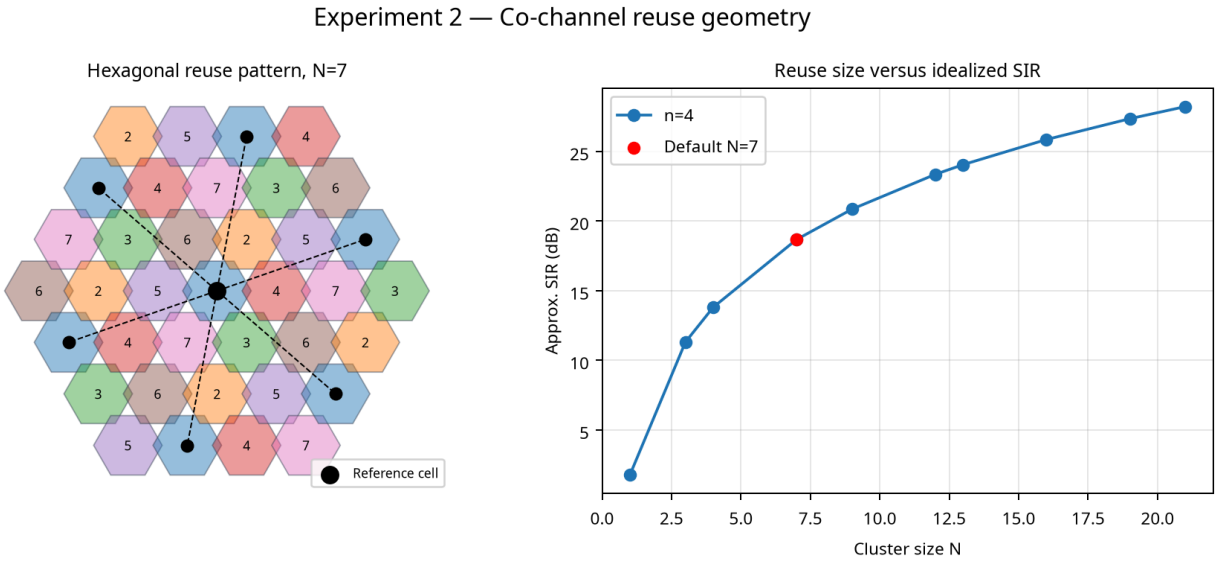


图 2 Hexagonal reuse schematic and idealized first-tier SIR versus cluster size.

2.5 Discussion

A larger cluster increases geometric isolation and the ideal SIR; however, it does not create additional spectrum. Select a cluster that satisfies the required interference target while preserving adequate channels per cell. The code takes the first valid (i, j) pair found in its bounded search for a selected N; cluster sizes must belong to the hexagonal sequence.

3 Study of Frequency Allocation and Channel Assignment in Cellular Networks

3.1 Aim and vocabulary

Convert an available per-direction band into discrete duplex channels, distribute channels among reuse groups, and visualize carrier assignment. A channel here is a paired uplink/downlink carrier; an actual air interface may subdivide carriers into time slots or other resources.

3.2 Channelization and allocation

For spectrum bandwidth B_{tot} in MHz per direction and channel spacing B_{ch} in kHz, the count used by the script is

$$M = \left\lfloor \frac{1000 B_{\mathrm{tot}}}{B_{\mathrm{ch}}} \right\rfloor.$$

With cluster size N, average channels per cell group are $\frac{M}{N}$. Sequential-block assignment gives contiguous channel-index blocks of nearly equal size; any remainder is assigned one at a time to the first groups. Interleaved assignment cycles group labels 1 through N, distributing adjacent channel indices among different groups. Interleaving in this demonstrator is only an assignment visualization and does not establish an interference-safe real-world channel plan.

For starting uplink frequency $f_{\mathrm{UL},0}$ and duplex spacing Δf, channel k (zero-based index) has

$$f_{\mathrm{UL,k}} = f_{\mathrm{UL},0} + k B_{\mathrm{ch}}, \quad f_{\mathrm{DL,k}} = f_{\mathrm{UL,k}} + \Delta f,$$

where B_{ch} must be expressed in MHz in the frequency increment. Uplink and downlink are paired to support frequency-division duplex operation.

3.3 Supplied MATLAB method

`exp3_frequency_allocation.m` validates that N has a valid lattice pair, calculates M, applies either sequential or interleaved allocation, and plots frequency-to-group mapping, group counts, a coloured hexagonal reuse pattern, and paired uplink/downlink channels.

3.4 Default worked result

Using 25 MHz per direction, a 200 kHz spacing (0.2 MHz), an 890 MHz uplink start, 45 MHz duplex spacing, and $N = 7$, the script forms $M = 125$ channel pairs. Six groups receive 18 channels and one receives 17; average allocation is $\frac{125}{7} = 17.857$. Uplink carriers run from 890.0 to 914.8 MHz in 0.2-MHz steps; their paired downlink range is 935.0 to 959.8 MHz. The values are an illustrative arithmetic plan: real band plans, guard bands, channel raster, regulatory assignments, and technology specifications must be checked before operational use.

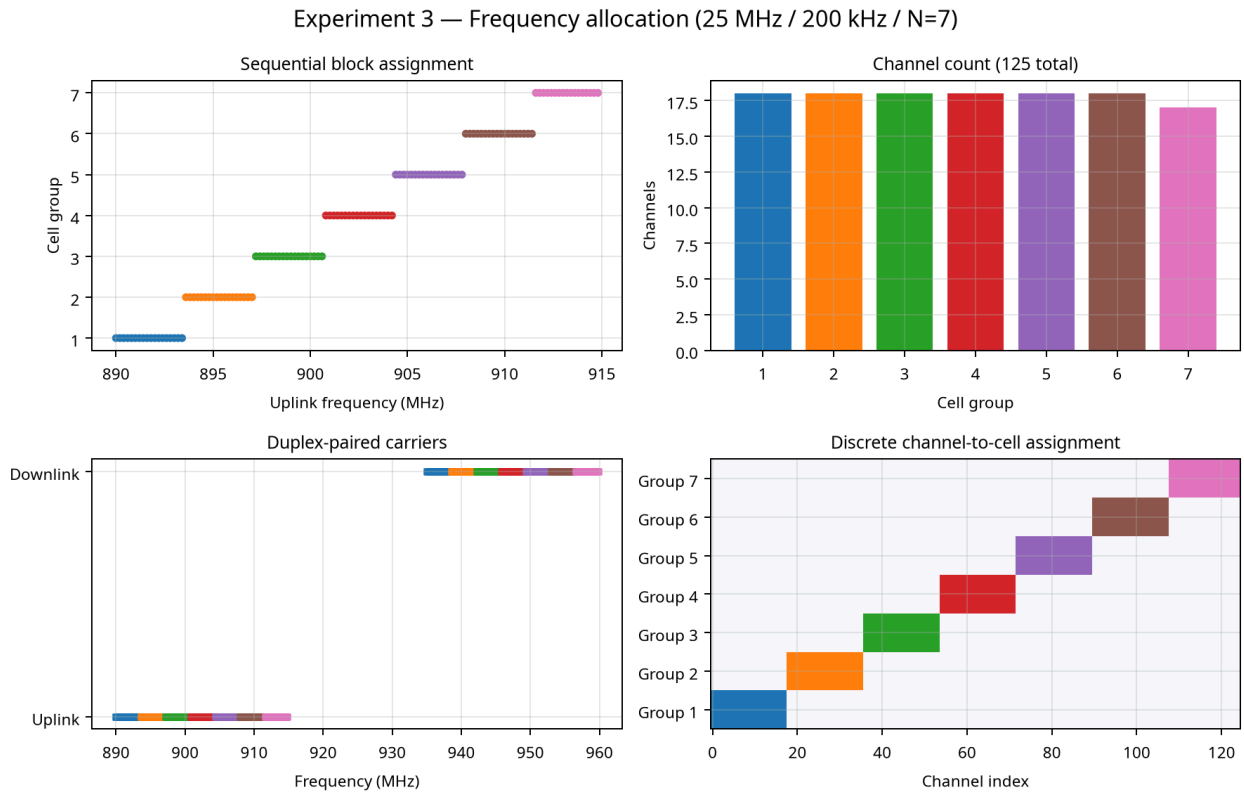


图 3 Sequential channel allocation, group channel counts, duplex pairing, and reuse-group schematic.

3.5 Interpretation

The integer remainder makes group capacities differ by at most one in sequential allocation. Equal count is not necessarily equal traffic capacity: subscriber distribution, busy-hour demand, interference constraints, and control-channel reservations are not modelled.

4 Simulation of GSM Call Setup, Location Update, and HLR/VLR Operations

4.1 Aim

Trace how GSM-style mobility management maintains subscriber reachability across location areas and how calls are routed using subscriber and serving-network records.

4.2 Network entities and records

The MS (mobile station) is identified by an IMSI (permanent subscriber identity) and addressed by an MSISDN (dialable number). The HLR (home location register) stores the subscriber's service profile and current serving-network/location information. Each visited MSC/VLR combines switching and a temporary local visitor record. BTS/BSC provide radio access and radio-resource control. A location area groups cells; a mobile normally reports a change of location area, not every cell reselection. For a mobile-terminated call, a GMSC queries the HLR and obtains routing information, commonly including a temporary MSRN (mobile station roaming number), to route the call to the serving MSC.

4.3 Location update sequence

When a registered mobile enters a different location area, the new serving MSC/VLR receives a location update request, identifies/authenticates the subscriber as needed, and contacts the HLR to update the serving VLR. The HLR requests cancellation at the old VLR and transfers subscriber data to the new VLR; acknowledgements complete registration, and the network may allocate a new TMSI (temporary identity). The conceptual transaction is MS → new VLR: Location Updating Request; new VLR → HLR: Update Location; HLR → old VLR: Cancel Location; HLR → new VLR: Insert Subscriber Data; new VLR → MS: Location Updating Accept. The implementation also represents radio channel access/release around this exchange.

4.4 Call setup logic represented in the code

For mobile-originated service, the MS requests access, the network establishes a signaling connection, performs authentication/ciphering as represented, receives the dialled number, assigns a traffic channel, and routes an ISUP call through the gateway toward the called party. For a mobile-terminated call, the GMSC queries the HLR using the dialled MSISDN; routing information and MSRN lead the call to the serving MSC, which pages the location area, obtains a response, assigns a traffic channel, and completes alerting/answer signaling. Exact signaling depends on release, services, and call state; the supplied sequence is pedagogical rather than a protocol conformance trace.

4.5 Supplied MATLAB method and visualizations

`exp4_gsm_hlr_vlr.m` creates synthetic IMSI/MSISDN strings, populates a simple HLR table and one VLR list per location area, moves a selected subscriber, and updates the old/new VLR membership. It draws a location-area map, a location-update message sequence when the serving area changes, and either an originating or terminating call sequence. Identifiers beginning with the code's example prefix are fabricated.

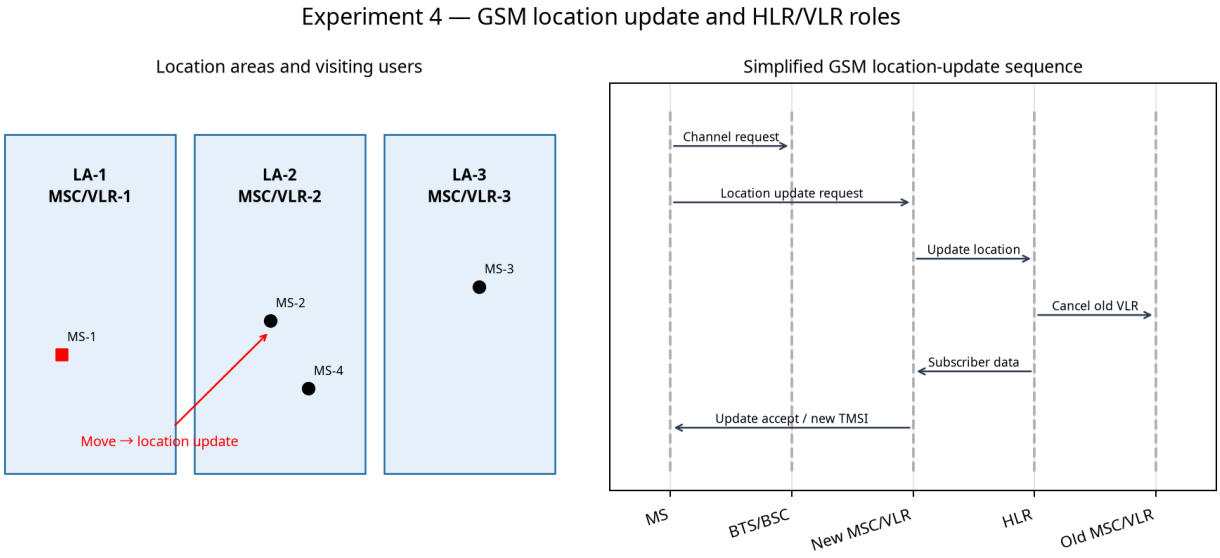


图 4 Location areas with a moving subscriber and a simplified location-update signaling sequence.

4.6 Observations and limitations

A stationary subscriber inside its current location area does not trigger an inter-area update in this model. A movement across a location-area boundary updates serving registration, after which paging for a terminating call targets the current location area. The MATLAB database is in-memory and simplified: it does not model authentication vectors, cryptographic security, actual radio channels, concurrent updates, database failures, or every standard information element. The graphic shows a shortened sequence for readability; consult the relevant GSM/3GPP procedures for implementation-level protocol details.

5 Simulation of Handover Between Two Cellular Cells

5.1 Aim

Compare strongest-signal handover with a hysteresis-based decision in the presence of distance-dependent path loss and correlated log-normal shadowing.

5.2 Received-signal model

For BS transmit power P_t in dBm, a reference path loss L_0 at d_0, and path-loss exponent n, the mean received level at distance d is

$$|P|_{r(d)} = P_t - L_0 - 10n \log_{10}\left(\frac{d}{d_0}\right).$$

The script calculates L_0 using a free-space anchor at $d_0 = 0.1$ km and the selected carrier. Shadowing is added in dB: $P_r = |P|_r + X$, where X is zero-mean Gaussian with standard deviation σ. Spatial correlation is approximated by a first-order process

$$X_k = aX_{k-1} + \sqrt{1-a^2}\sigma Z_k, \quad a = \exp\left(-\Delta\frac{s}{d_c}\right),$$

where Z_k is independent standard normal, $\Delta s = vT_s$, and d_c is decorrelation distance. This yields variance σ^2 in steady state.

5.3 Handover logic

Without hysteresis, the serving cell is whichever has greater sampled RSS; rapid shadowing fluctuations around equality can cause ping-pong handovers. With hysteresis H, a mobile on BS1 changes to BS2 only if $P_{\mathrm{r,2)>P_}}r, 1) + H$; on BS2 it returns only if $P_{\mathrm{r,1)>P_}}r, 2) + H$. A minimum usable level $S_{\min}$ separately identifies outage:

$$P_{\text{serving}} < S_{\min}.$$

Hysteresis lowers unnecessary switching but can delay a switch, so it is not guaranteed to reduce all outage.

5.4 Supplied MATLAB method

`exp5_handover.m` moves a mobile across a two-BS line, synthesizes correlated shadowing using the recurrence above, compares no-margin and margin-based serving-cell decisions, and counts handovers and samples below sensitivity. It plots the two signal traces, serving-cell state, and selected serving RSS.

5.5 Default interpretation

Defaults are $D = 2$ km, $f_c = 900$ MHz, $P_t = 43$ dBm, $n = 3.5$, $\sigma = 4$ dB, $d_c = 50$ m, $H = 5$ dB, $S_{\min} = -100$ dBm, $v = 60$ km/h, and $T_s = 0.5$ s. The ideal symmetric mean-power crossing is at $\frac{D}{2} = 1$ km. Sampled shadowing shifts actual switching events; hysteresis prevents a return until the signal advantage exceeds 5 dB. Since the exact random stream and sampling can differ across MATLAB and Python, event counts should be taken from the student's MATLAB run, not inferred from the illustration.

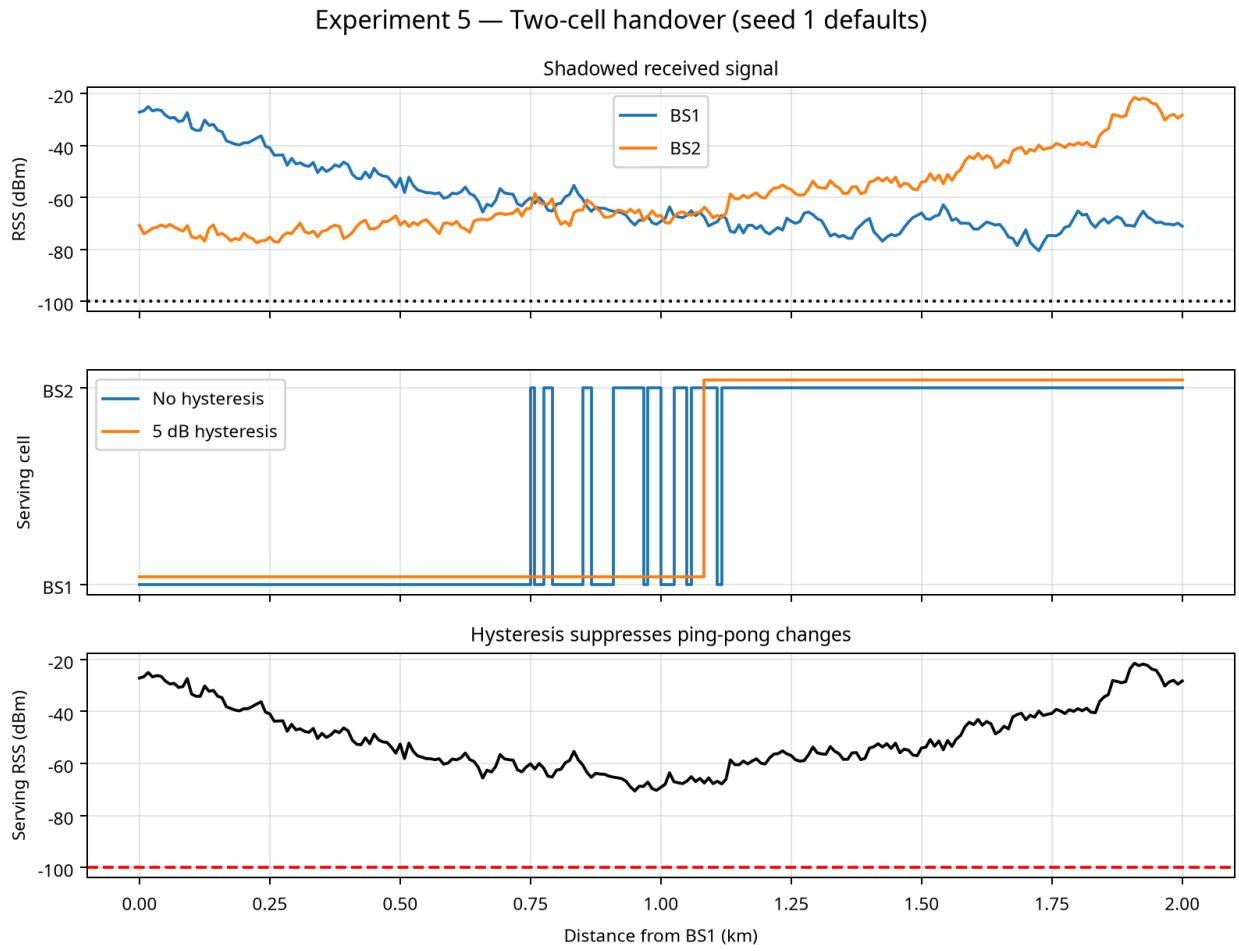


图 5 Correlated-shadowing RSS traces, serving-cell decisions, and serving signal level.

5.6 Limitations

This is a one-dimensional, two-cell, single-mobile model. It does not include interference, cell load, measurement filtering/quantization, handover execution latency, neighbor lists, hysteresis timers, failure probabilities, or a radio link-control state machine. The Gaussian shadowing process is spatially correlated but independently generated for each base station.

6 Received Signal Strength-Based Base Station Selection

6.1 Aim

Select the strongest base station at a mobile location and estimate best-server regions and coverage above a receiver threshold over a two-dimensional area.

6.2 Propagation and selection rule

For base station b at location (x_b, y_b), mobile position (x, y), distance

$$d_b = \sqrt{(x - x_b)^2 + (y - y_b)^2},$$

the code's mean log-distance model is

$$P_{\mathrm{r,b}} = P_{\mathrm{t,b}} - L_0 - 10n \log_{10}\left(\frac{d_b}{d_0}\right).$$

At the selected mobile position, it adds independent shadowing σZ_b and selects

$$b^* = \arg\max_b P_{\mathrm{r,b}}.$$

Service is available when the strongest received power is at least $S_{\min}$. For the grid map, the supplied script uses deterministic mean RSS only; its service-area percentage is the fraction of sampled grid points whose strongest mean RSS exceeds sensitivity. The area served by BS b is the fraction of all grid points that both lie above threshold and have b as their strongest server.

6.3 Supplied MATLAB method

`exp6_rss_bs_selection.m` sets a square service region, default BS coordinates for up to four stations, evaluates a 200-by-200 RSS grid for each station, and creates (i) a strongest-server map, (ii) maximum-RSS heat map with sensitivity contour, (iii) RSS bars at the mobile, and (iv) per-BS share bars. The MS-position RSS includes a seeded shadowing sample; the spatial grid does not.

6.4 Default interpretation

The defaults use a 6 km square, four BSs at fractional positions of the square, $P_t = 43$ dBm, $f_c = 900$ MHz, $n = 3.5$, $\sigma = 4$ dB, and $S_{\min} = -100$ dBm. Strongest-RSS boundaries lie approximately where competing mean powers are equal, not necessarily at geometric Voronoi boundaries when transmit powers differ. The sensitivity contour separates estimated coverage from no service. The finite grid makes area shares approximate and resolution-dependent.

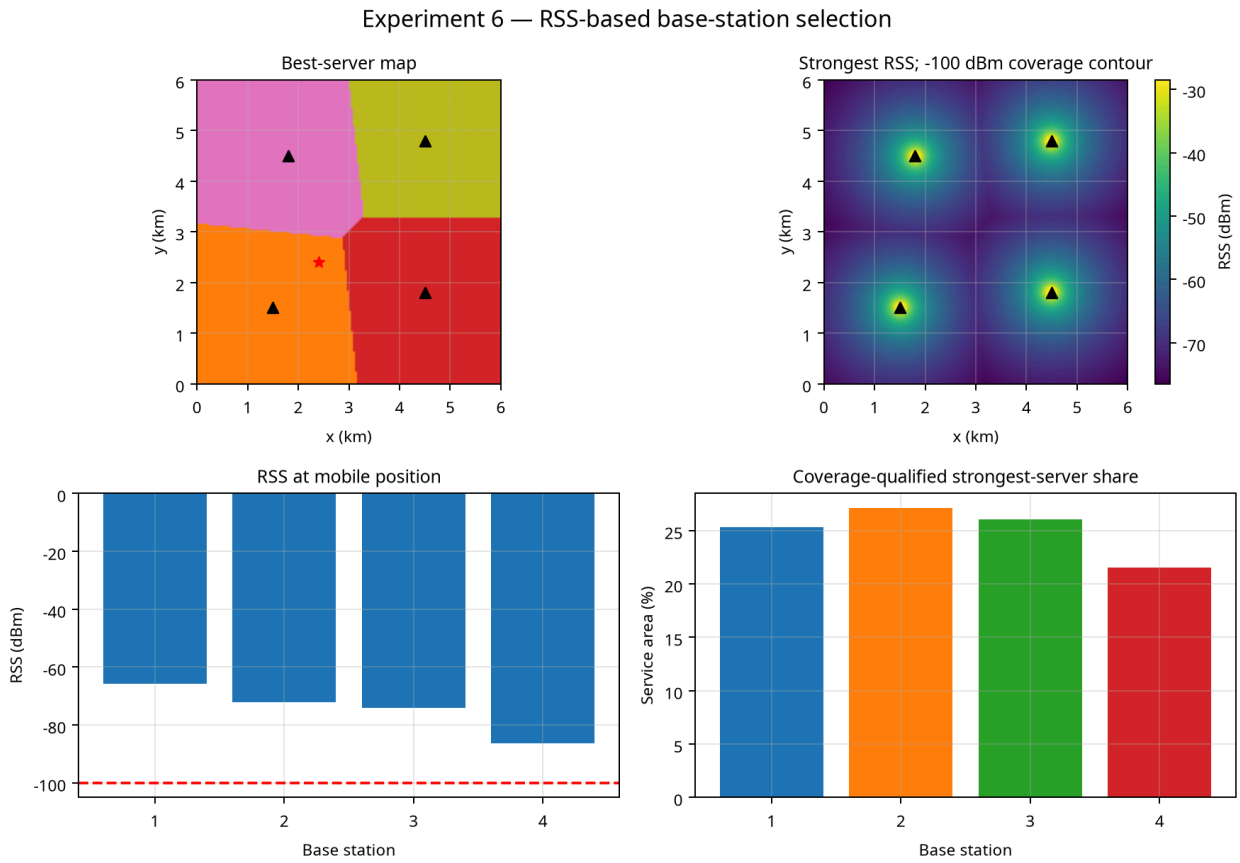


图 6 Illustrative best-server regions, sensitivity contour, and RSS at the sample mobile location.

6.5 Limitations

Strongest RSS is not always the best user experience: a practical scheduler considers interference, load, quality, mobility state, and technology-specific metrics. Independent shadowing at one point does not model correlated spatial shadowing across the map. This experiment is a received-power selection demonstrator, not a complete handover or cell-selection procedure.

7 Free-Space Path Loss and Received Power Analysis Using MATLAB

7.1 Aim

Quantify ideal free-space attenuation, received power, link margin, and a threshold-limited range as functions of distance, carrier frequency, transmit power, antenna gains, and system loss.

7.2 Friis transmission and decibel form

For unobstructed far-field propagation between matched isotropic antennas,

$$\frac{P_r}{P_t} = G_t G_r \left(\frac{\lambda}{4\pi d}\right)^2.$$

The free-space path loss is

$$L = \left(4\pi\frac{d}{\lambda}\right)^2, \quad L_{\mathrm{dB}} = 20\log_{10}\left(4\pi\frac{d}{\lambda}\right).$$

For distance d in km and frequency f in MHz, the common equivalent is

$$L_{\mathrm{dB}} = 32.44 + 20\log_{10}(d_{\mathrm{km}}) + 20\log_{10}(f_{\mathrm{MHz}}).$$

The budget is

$$P_r = P_t + G_t + G_r - L_s - L,$$

where gains and system losses are in dB. Convert to watts with $P_{\mathrm{W}} = 10^{\frac{P_{\mathrm{dBm}}-30}{10}}$. Link margin is $M = P_r - S_{\mathrm{min}}$; setting it to zero gives

$$d_{\mathrm{max}} = 10^{\frac{P_t+G_t+G_r-L_s-S_{\mathrm{min}}-32.44-20\log_{10}(f_{\mathrm{MHz}})}{20}}.$$

7.3 Supplied MATLAB method

`exp7_free_space_path_loss.m` compares the engineering FSPL expression to the exact Friis wavelength form at a requested point, calculates power and threshold range, then plots loss versus distance and frequency, power versus distance, and maximum range versus transmitter power.

7.4 Default worked result

For $f_c = 900$ MHz, $P_t = 43$ dBm, zero gains/losses, $d = 5$ km, and $S_{\mathrm{min}} = -100$ dBm, $\lambda = 0.3333$ m; $L = 105.50$ dB; $P_r = -62.50$ dBm, about 5.62×10^{-10} W; and link margin is about 37.50 dB. The formula gives a free-space sensitivity range near 374.8 km. That large value is a mathematical extrapolation; Earth curvature, obstruction, Fresnel clearance, atmospheric effects, interference, practical cell-planning limits, and the validity of the free-space model are omitted.

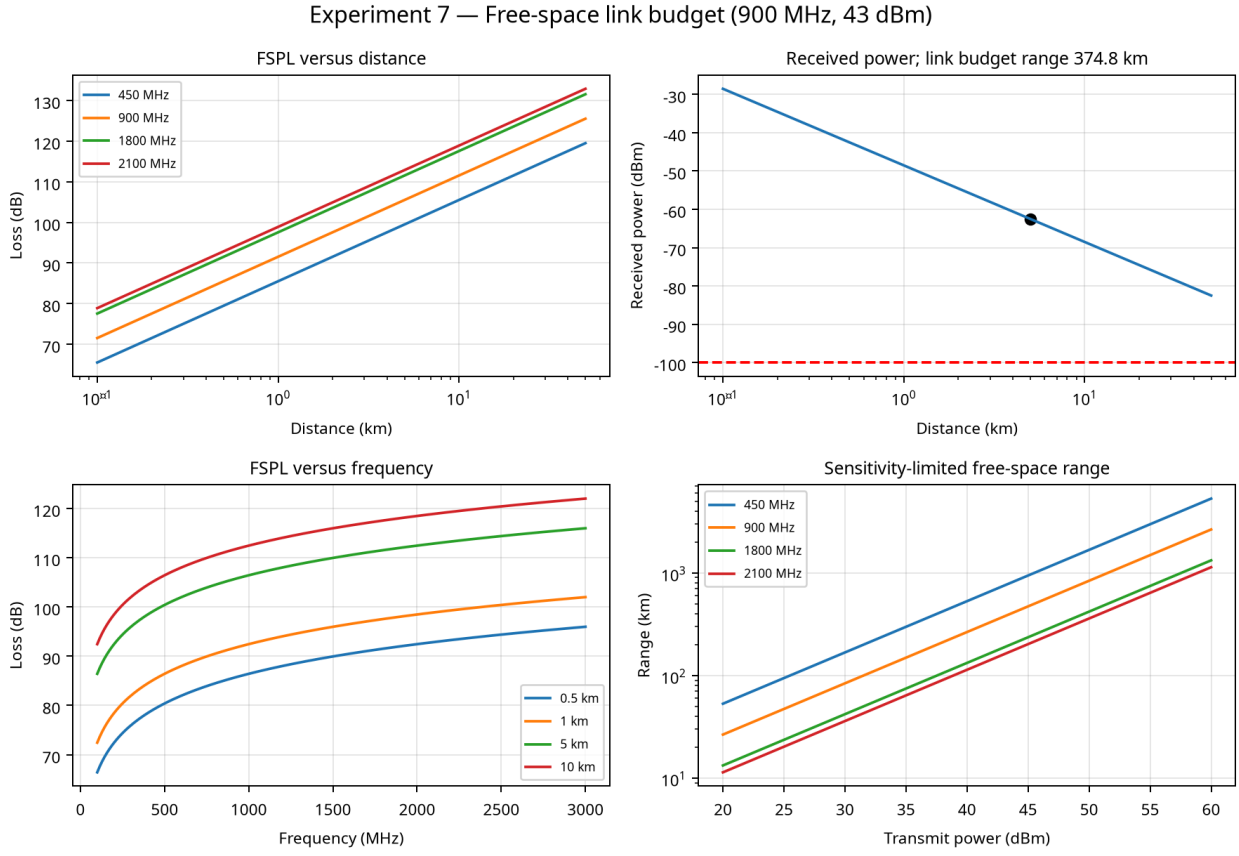


图 7 Free-space attenuation and ideal threshold-limited link-budget trends.

7.5 Interpretation and checks

Doubling distance or frequency adds approximately 6.02 dB of free-space loss. Adding 1 dB of transmitter power or antenna gain adds 1 dB of received power, all else equal. The code assumes the conventional 32.44 constant for the specified km/MHz units; its separately computed exact formula converts km to metres and frequency to hertz.

8 Analysis of Cell Coverage, Cell Radius, and Cell Splitting

8.1 Aim

Relate a smaller cell radius to coverage footprint, transmit-power adjustment, cell count, and nominal system channel capacity.

8.2 Hexagonal area and split scaling

For a regular hexagon with center-to-vertex radius R, area is

$$A_{\text{cell}} = \left(3\frac{\sqrt{3}}{2}\right)R^2.$$

If cell radius is reduced by splitting factor $S > 1$, then $R' = \frac{R}{S}$ and

$$A_2 = \frac{A_1}{S^2}, \quad \frac{A_1}{A_2} = S^2.$$

Thus, for a fixed service area and unchanged cluster allocation, idealized cell density—and nominal channels per area—can rise approximately as S^2. This assumes sufficient sites, backhaul, control resources, and interference coordination.

For the code's log-distance model, keeping a similar received-power boundary at radius $\frac{R}{S}$ requires a lower transmit power:

$$P_2 = P_1 - 10n\log_{10}(S).$$

This relation changes the link-budget coverage contour; it is distinct from simply multiplying channel capacity by S^2.

8.3 Supplied MATLAB method and results

`exp8_cell_splitting.m` accepts a radius or derives one from the threshold link budget, applies the splitting factor, estimates hex-cell area and the number of cell centers in a finite square, allocates $\frac{C_{\text{tot}}}{N}$ channels per cell, and compares pre/post cell layouts, link curves, and estimated capacity. If the radius is omitted, the code derives $R = d_0 10^{\frac{P_t - L_0 - S_{\min}}{10n}}$ with $d_0 = 0.1$ km.

For the code defaults $f_c = 900$ MHz, $P_t = 43$ dBm, $S_{\min} = -100$ dBm, $n = 3.5$, $d_0 = 0.1$ km, $S = 2$, $N = 7$, and $C_{\text{tot}} = 280$, $L_0 = 71.52$ dB and the derived R is approximately 11.02 km. Hence $R' = 5.51$ km; $A_{\text{cell}} \approx 315.7$ km² before and one quarter after; the transmit power reduction is $10(3.5)\log_{10}(2) = 10.54$ dB, so $P_t' \approx 32.46$ dBm. The theoretical channel allocation is 40 channels per cell. The script's actual system totals depend on its grid-counting rule and service-region boundary.

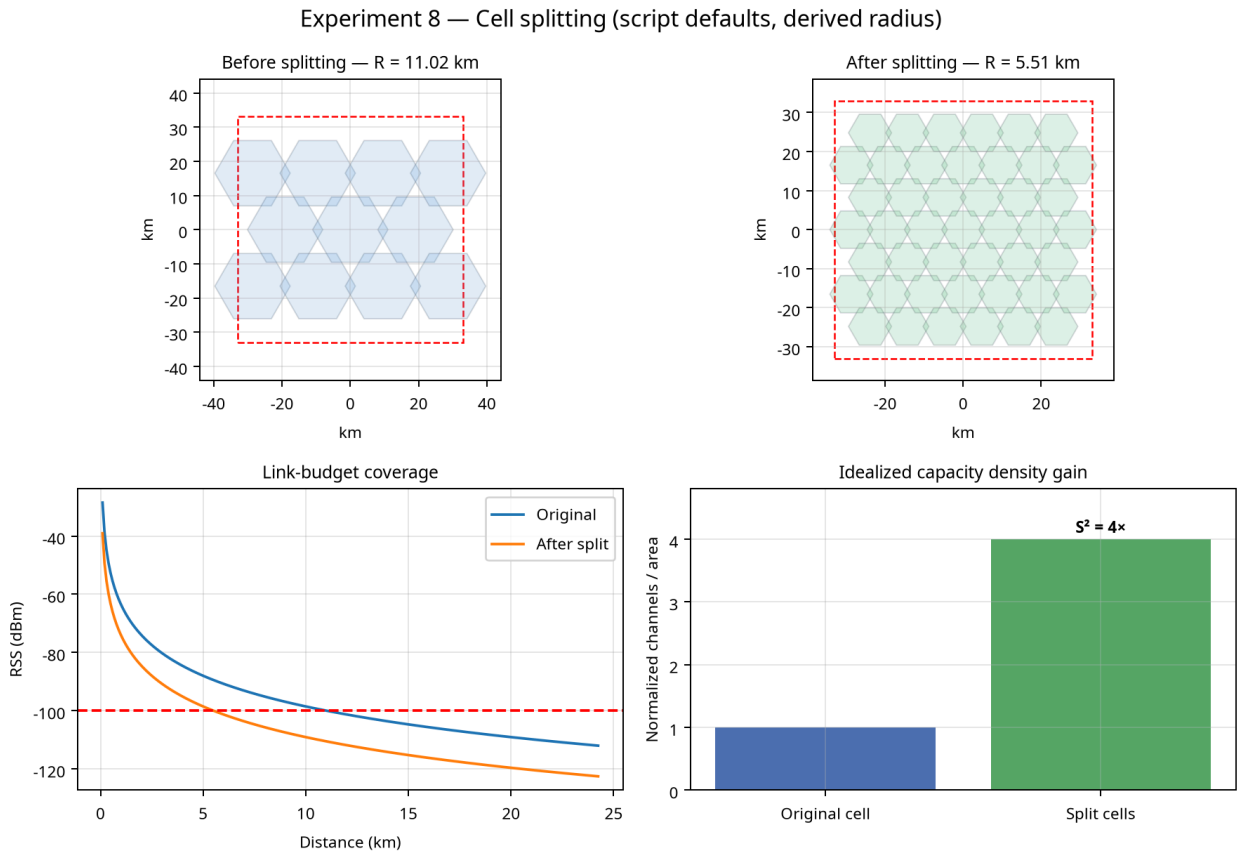


图 8 Cell-radius and link-budget changes under a factor-two split; capacity density is idealized.

8.4 Discussion and limitations

Cell splitting increases the number of low-power sites required to cover a fixed region; the plotted finite grid and cell-center counting are a geometric approximation, not an optimized radio plan. Boundary cells are counted by center inclusion, so small changes in service-area dimensions can change the integer total. Capacity gains assume every new cell can reuse the same cluster allocation while maintaining acceptable co-channel interference. The default link-budget-derived radius is notably large because the model omits practical range constraints; instructors may enter a more realistic study radius.

9 Cellular Traffic, Channel Capacity, and Erlang-B Blocking Probability Analysis

9.1 Aim and traffic definitions

Estimate circuit-switched call blocking, carried traffic, channel utilization, channel count for a target grade of service, and maximum subscriber count for a cell.

For U subscribers, per-subscriber call arrival rate λ calls/hour, and mean holding time h seconds, the offered traffic per user and per cell are

$$A_u = \lambda \frac{h}{3600} \quad A = U A_u.$$

An Erlang represents one continuously occupied circuit on average. Offered load includes traffic that may ultimately be blocked.

9.2 Erlang-B formula and stable recurrence

For C identical trunks, Poisson arrivals, independent call holding times, and no queue (blocked calls are cleared), the Erlang-B loss probability is

$$B(A,C) = \frac{\frac{A^C}{C!}}{\sum_{k=0}^{C} \frac{A^k}{k!}}.$$

A numerically stable recursive form is

$$B(A,0) = 1, \quad B(A,C) = \frac{AB(A,C-1)}{C + AB(A,C-1)}.$$

Carried traffic is $A_c = A(1-B)$; mean channel utilization is $\frac{A_c}{C}$. Blocked calls per hour are approximately $U\lambda B$ when λ is per user. The required channel count is the smallest integer C with $B(A,C) \le g$. Conversely, maximum traffic for fixed C and target GoS solves $B(A,C) = g$; the script uses binary search. Approximate maximum subscriber count is $\left\lfloor \frac{A_{\max}}{A_u} \right\rfloor$.

9.3 Supplied MATLAB method

`exp9_erlang_b.m` computes theoretical blocking using the recurrence, then simulates a finite Poisson arrival stream with exponential interarrival times and exponential holding times. An arrival is blocked if all C channels remain occupied. The random simulation estimates blocking by $P_{\text{sim}} = \frac{N_{\text{blocked}}}{N_{\text{calls}}}$. Its four plots compare blocking against offered load and channels, channel capacity at several target GoS values, and offered versus carried traffic.

9.4 Default worked result

Defaults are $C = 20$, $U = 400$, $\lambda = 1$ call/hour/user, $h = 120$ s, and target $g = 0.02$. Thus $A_u = 0.03333$ E and $A = 13.333$ E. Erlang-B gives $B \approx 0.02167$ (2.17%), slightly above 2%. Carried load is $13.333(1 - 0.02167) \approx 13.044$ E and average utilization about 65.2%. The smallest channel count that meets 2% is 21; at 20 channels, the largest offered load meeting 2% is about 13.18 E, corresponding to 395 users at the stated traffic per user. The simulated blocking ratio should fluctuate around theoretical blocking; with 100,000 calls its sampling uncertainty is on the order of $\sqrt{\frac{B(1-B)}{N_{\text{calls}}}}$.

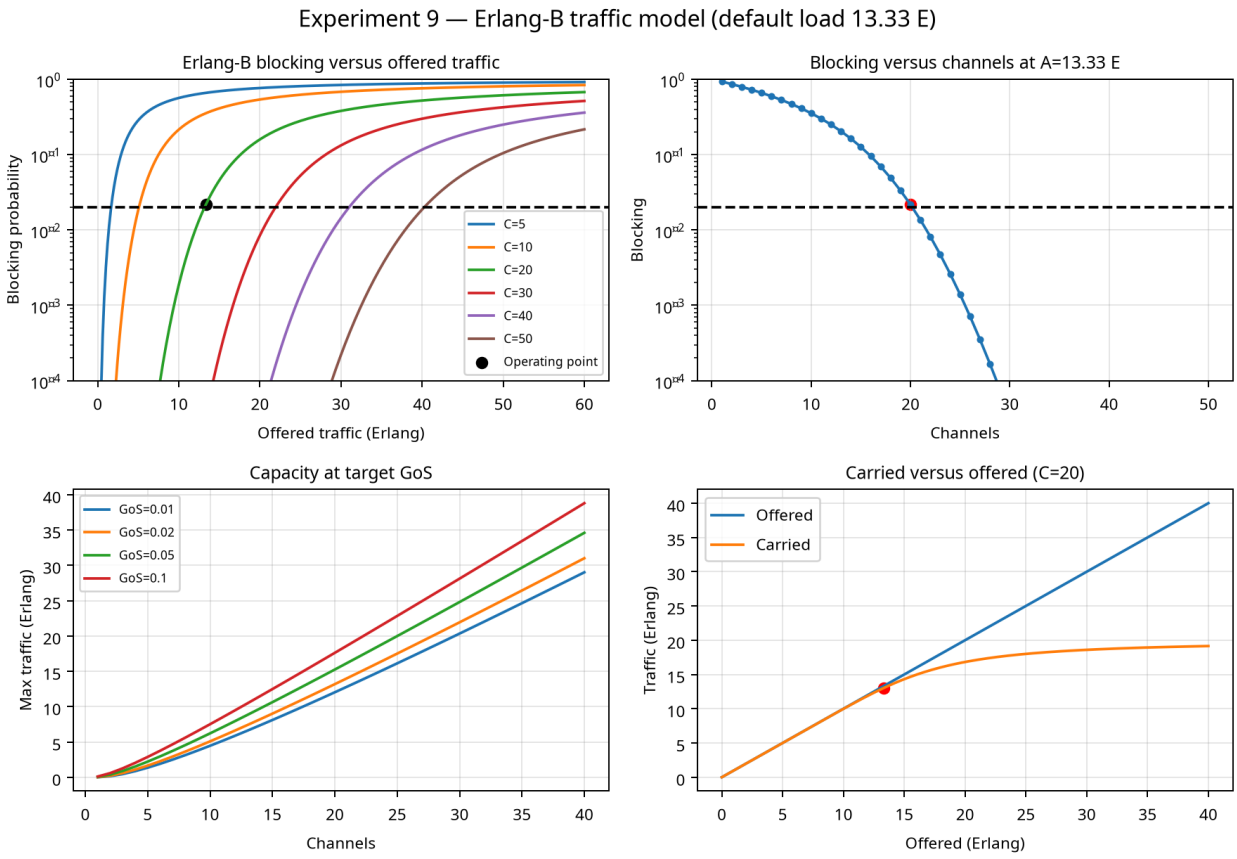


图 9 Erlang-B blocking, channel requirement, traffic capacity, and carried/offered traffic.

9.5 Interpretation and limitations

Adding channels reduces blocking for a fixed offered load; increasing load raises blocking. There is no queue: Erlang-B applies to a loss system rather than a waiting system. The simulation assumes a stationary homogeneous Poisson arrival process and exponential holding duration; it does not model retries, user classes, priority, correlated busy-hour demand, handover traffic, or reserved channels. Finite-run estimates need confidence intervals and repeat seeds; they should not be expected to equal the theoretical value exactly.

10 Overall Conclusions

The nine exercises jointly show how mobility, geometry, radio propagation, resource assignment, and traffic demand constrain cellular service. Doppler depends on radial speed and carrier frequency; reuse geometry trades spectrum fraction against idealized interference; channel allocation maps finite spectrum to cells; HLR/VLR procedures preserve reachability as a subscriber moves; hysteresis stabilizes handover; received-power maps estimate best-server coverage; FSPL offers a baseline link budget; cell splitting raises idealized spatial capacity at the cost of added sites; and Erlang-B quantifies loss under finite circuit capacity. The numerical outputs are only as credible as their assumptions. Real networks additionally require measured propagation, regulatory frequency plans, interference and load models, standardized signaling, engineering margins, and validated traffic data.

11 Suggested viva questions

1. Why does maximum Doppler scale linearly with speed and carrier frequency, and what is the effect of arrival angle?
2. Which integer pairs produce valid cluster sizes, and what assumptions are hidden in the six-interferer SIR equation?
3. Why can interleaved assignment not, by itself, guarantee an interference-safe frequency plan?
4. What information is held conceptually in HLR versus VLR, and why is paging restricted to a location area?
5. How does hysteresis suppress ping-pong handovers, and what risk follows from making it too large?
6. Why does a strongest-RSS server map differ from a load-aware selection map?
7. Why is a very long range calculated by a free-space budget not a practical cell-radius prediction?
8. Under what ideal assumptions does cell splitting yield an S^2 capacity-density gain?
9. Why is Erlang-B appropriate for a no-queue cellular circuit group, and how should Monte Carlo error be reported?

12 References and model provenance

The equations and conventions in this explainer are standard engineering models: Doppler projection and the Jakes coherence-time approximation; hexagonal frequency-reuse geometry; Friis free-space transmission and log-distance shadowing; circuit-switched Erlang loss traffic. Protocol descriptions are intentionally conceptual and follow established GSM architecture terms (MS, BTS/BSC, MSC/VLR, HLR, GMSC, IMSI, TMSI, MSISDN, and MSRN). No outside measurements or unprovided network configuration were assumed. For a formal submission, add the institution's prescribed textbook/standard citations, course metadata, student names, instructor, date of execution, and the MATLAB-generated output figures.